

WEATHER DATA ANALYSIS

Executive Report | 2022-2024

OBSERVATIONS	6,576
CITIES	6
MEAN TEMPERATURE	25.8 C
EXTREME INDICATORS	904

Executive Summary

This report analyses 6,576 daily city observations across 6 Indian cities from 2022-01-01 to 2024-12-31. Chennai has the highest mean temperature (28.0 C), while Chennai records the greatest accumulated rainfall.

The dataset contains 439 heatwave indicators, 542 heavy-rain indicators, and 12 cold-day indicators. These are portfolio thresholds rather than official city warnings.

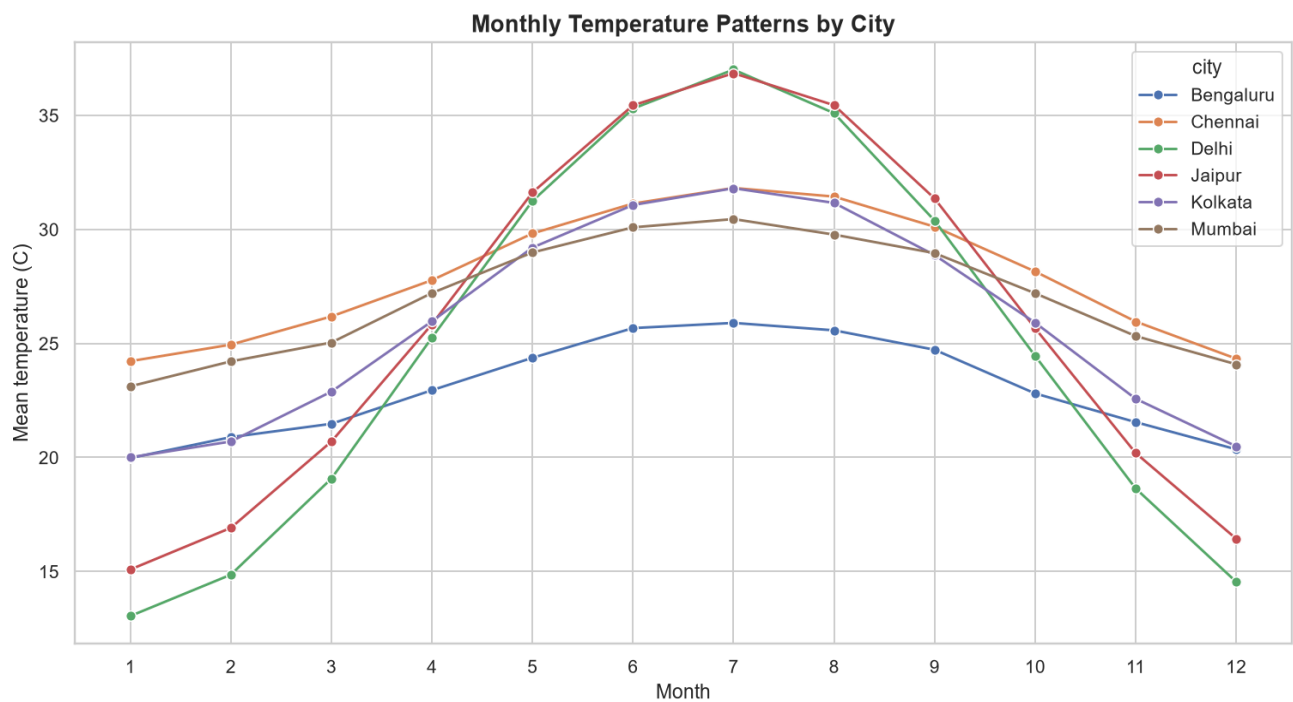
1. Methodology

The pipeline validates schema, duplicates, dates, measurement ranges, and minimum/maximum temperature ordering. Missing rainfall, humidity, and wind observations are imputed with city-month medians. Calendar, season, temperature range, and four extreme-event flags are engineered.

Statistical Summary

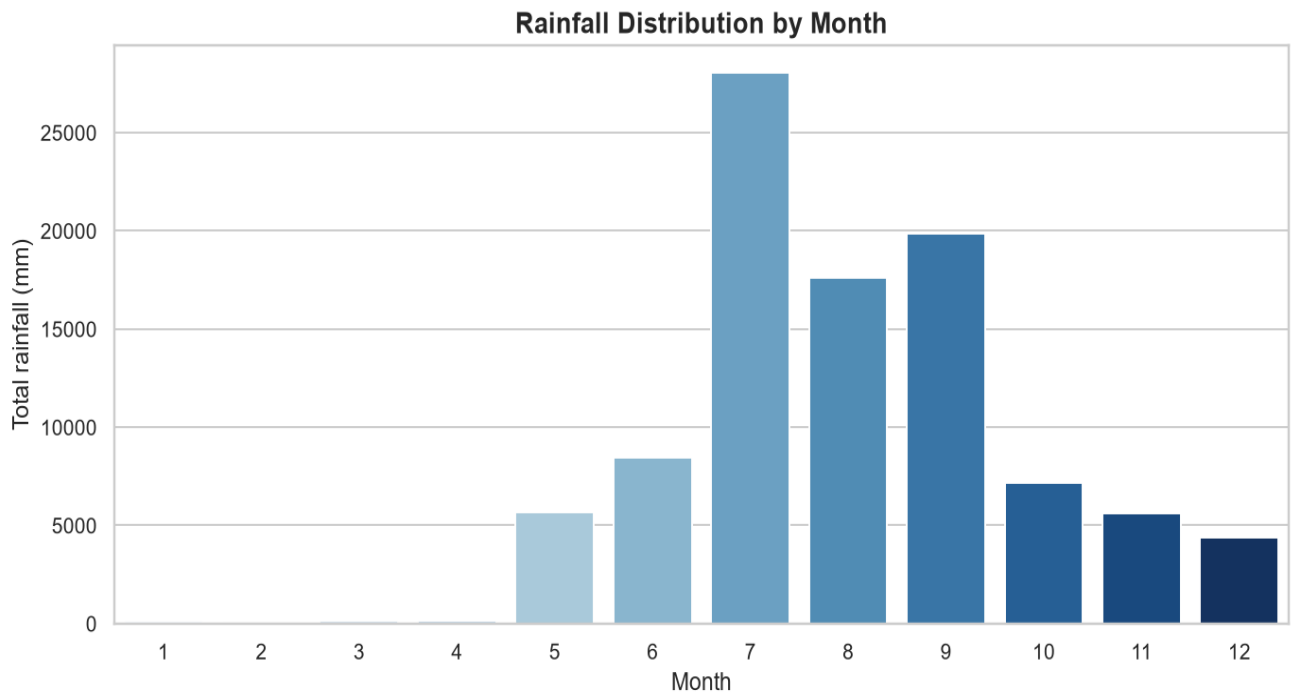
Metric	Value
Median temperature	25.80 C
Rainy observations	1,977
Maximum daily rainfall	333.7 mm
Rainfall-humidity correlation	0.536
Temperature-humidity correlation	0.300

2. Temperature Trends



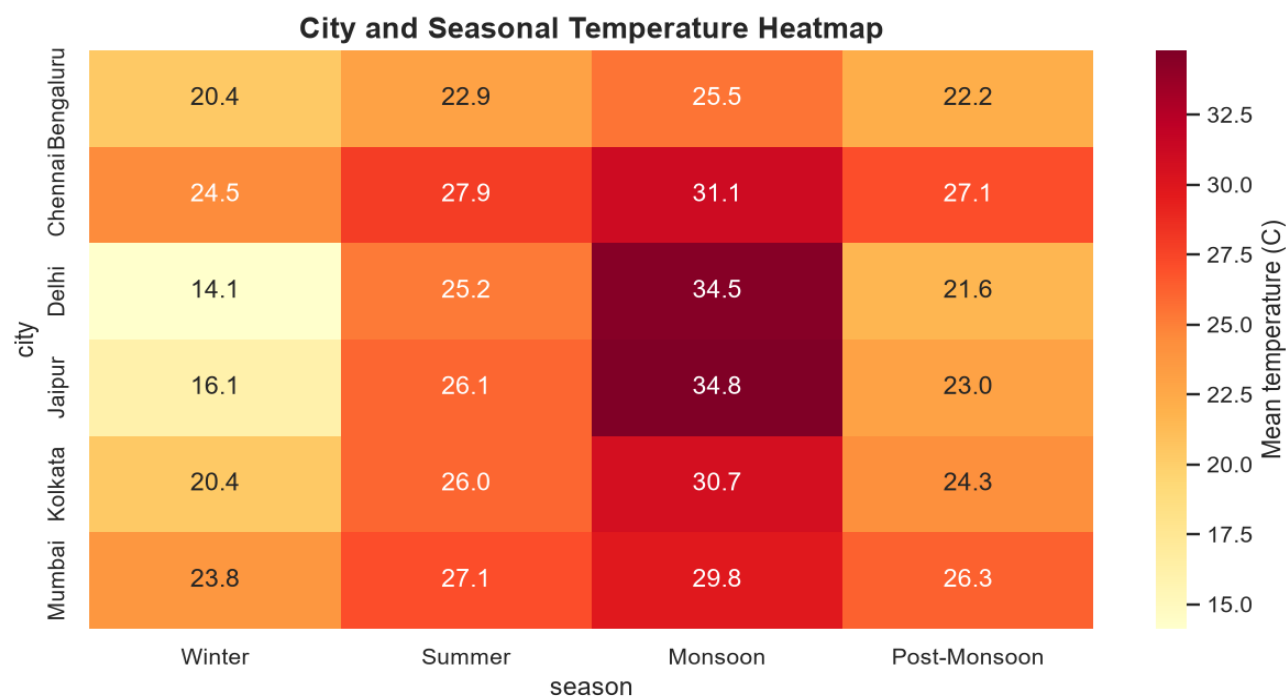
City lines show distinct annual cycles and why local seasonal baselines are more useful than a single portfolio average.

3. Rainfall Distribution



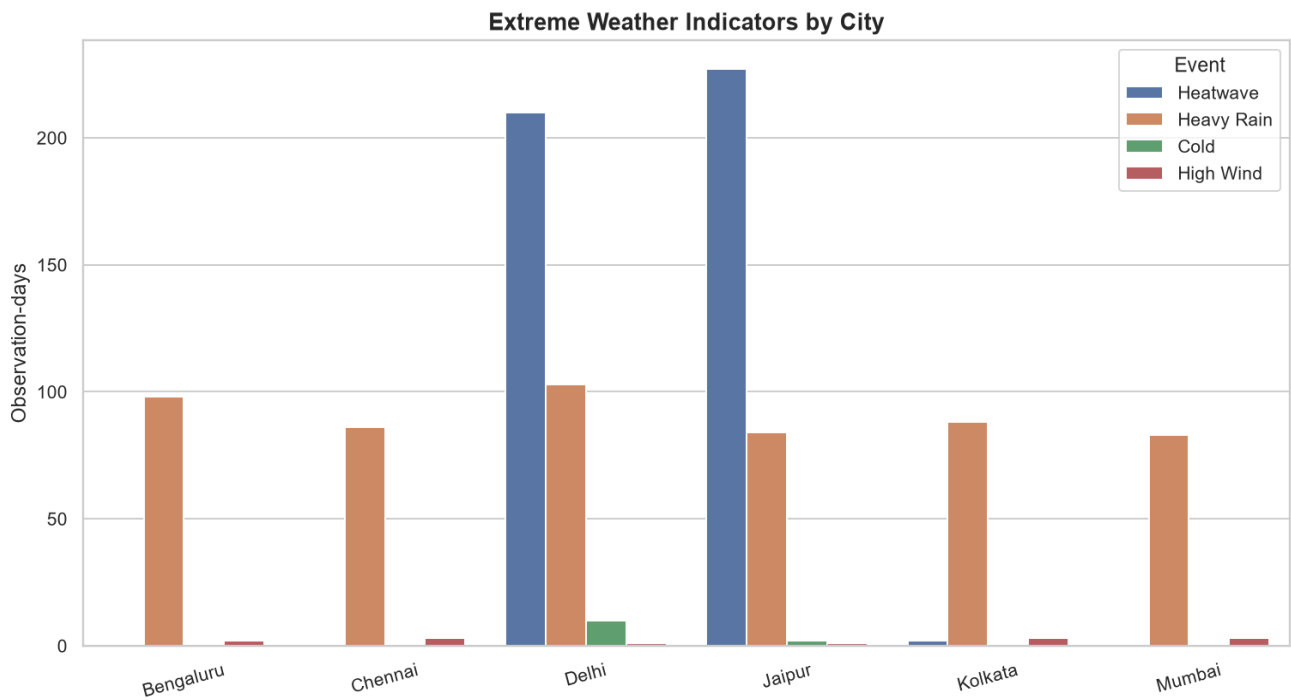
Rainfall is concentrated in a limited set of months, supporting seasonal drainage, supply, and staffing preparation.

4. Seasonal Patterns



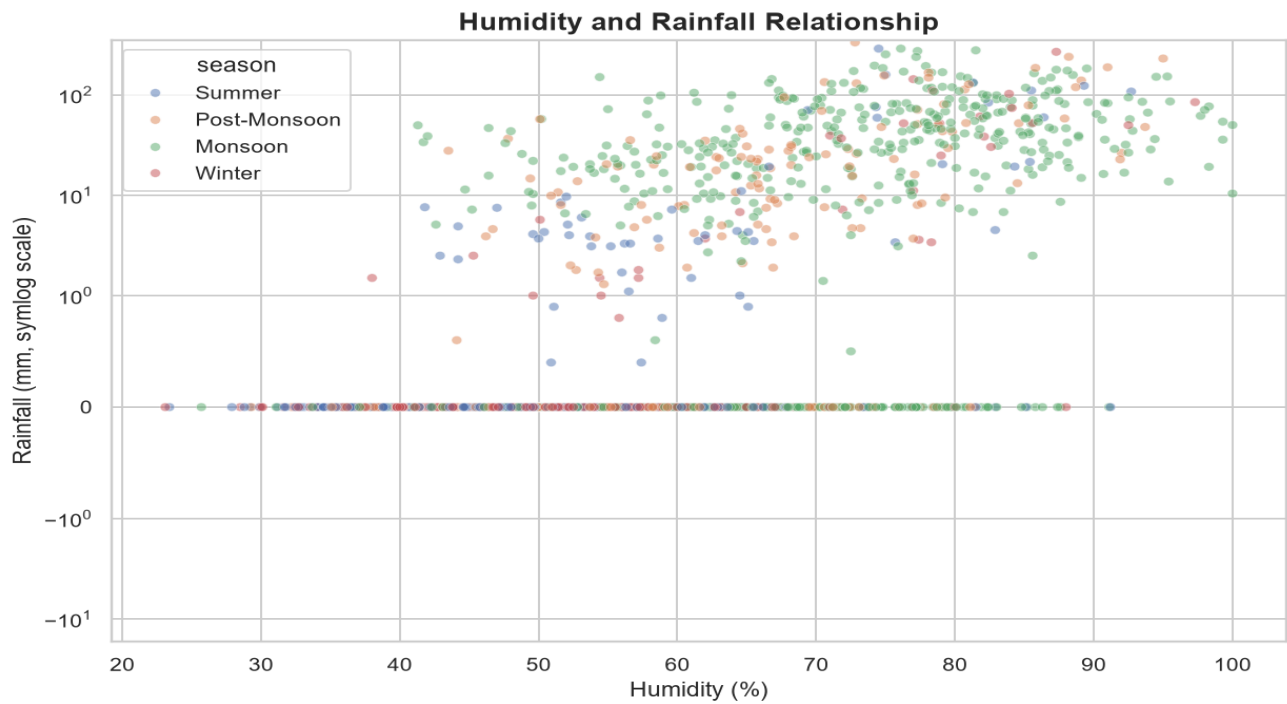
The heatmap makes city-season differences explicit and highlights locations with relatively stable or volatile temperatures.

5. Extreme Weather Indicators



Counts compare analytical heat, heavy-rain, cold, and high-wind indicators. They measure exposure in this simulated dataset, not official disaster declarations.

6. Humidity and Rainfall



The scatter plot shows how humidity and rainfall coexist across seasons; the nonlinear distribution and many dry days limit simple interpretation.

7. Recommendations and Limitations

Seasonal readiness: Build separate city calendars for heat, rainfall, and maintenance preparations.

Drainage planning: Inspect drainage and emergency supplies before the highest-rainfall months.

Heat preparedness: Trigger communication and staffing reviews as forecasts approach 40 C.

Local thresholds: Avoid applying one threshold or portfolio average to every city.

Operational validation: Replace simulated observations with verified meteorological data and official local definitions before real decisions.

Limitations

This dataset is simulated for portfolio practice and is not an official weather record. The extreme-event thresholds are simplified analytical rules. Correlations are descriptive, and three years is too short for climate-change attribution.